

# Module

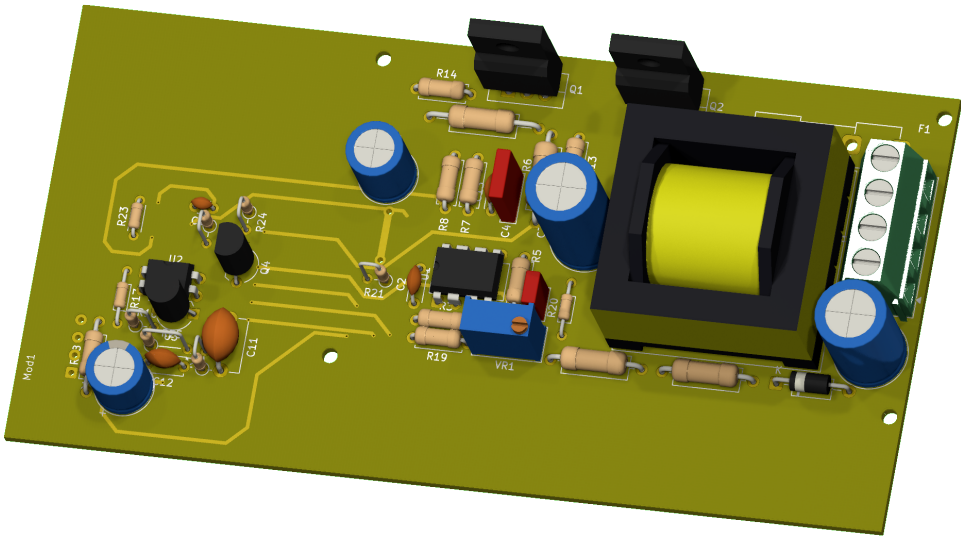
Variant: RELEASED

Rev 1.0.0

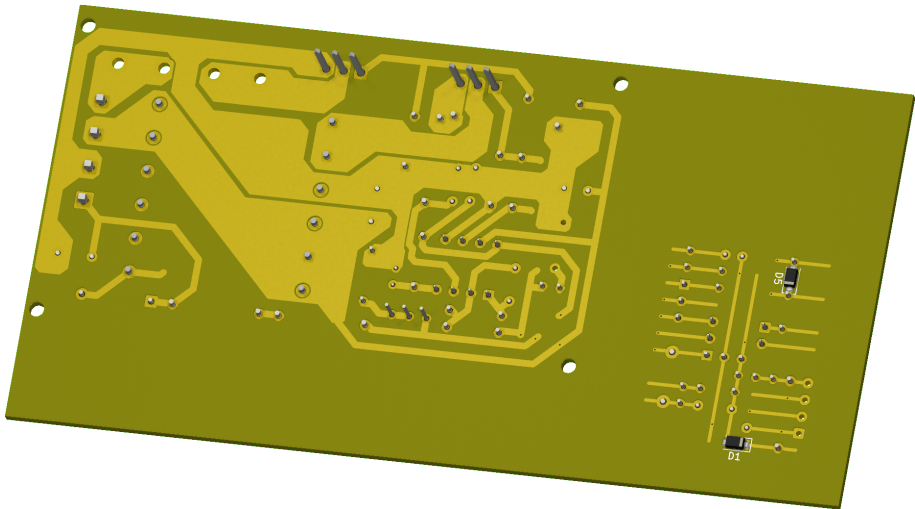
2026-01-24

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TOP VIEW



BOTTOM VIEW



DRAFT - Early concept stage, schematic drafted.

PRELIMINARY - Minor schematic changes are expected, ready for layout planning.

CHECKED - DRC completed, schematic and layout ready for review.

RELEASED - Design ready for manufacturing.

## NOTE STYLING CONVENTION

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

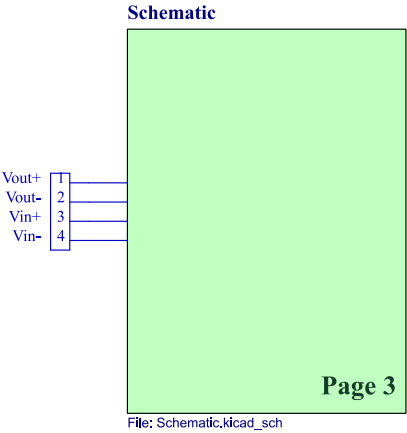
Example text for critical layout guidelines.

Date: 24-Jan-2026

|  |                  |                                           |                 |                                            |                      |
|--|------------------|-------------------------------------------|-----------------|--------------------------------------------|----------------------|
|  | Comments:        | Company:<br>FR                            |                 | Variant:<br>RELEASED                       | Git Hash:<br>f8fc805 |
|  | Sheet Title:     | Board Name:<br>Module                     |                 | Project Name:<br>Isolated step-up 12:450 V |                      |
|  | Sheet Path:<br>/ | File Name:<br>Isolated_12to450V.kicad_sch | Designer:<br>FR | Date:<br>2025-12-20                        | Revision:<br>1.0.0   |
|  |                  | Reviewer:<br>NA                           | Size:<br>A3     | Sheet:<br>1 of 4                           |                      |

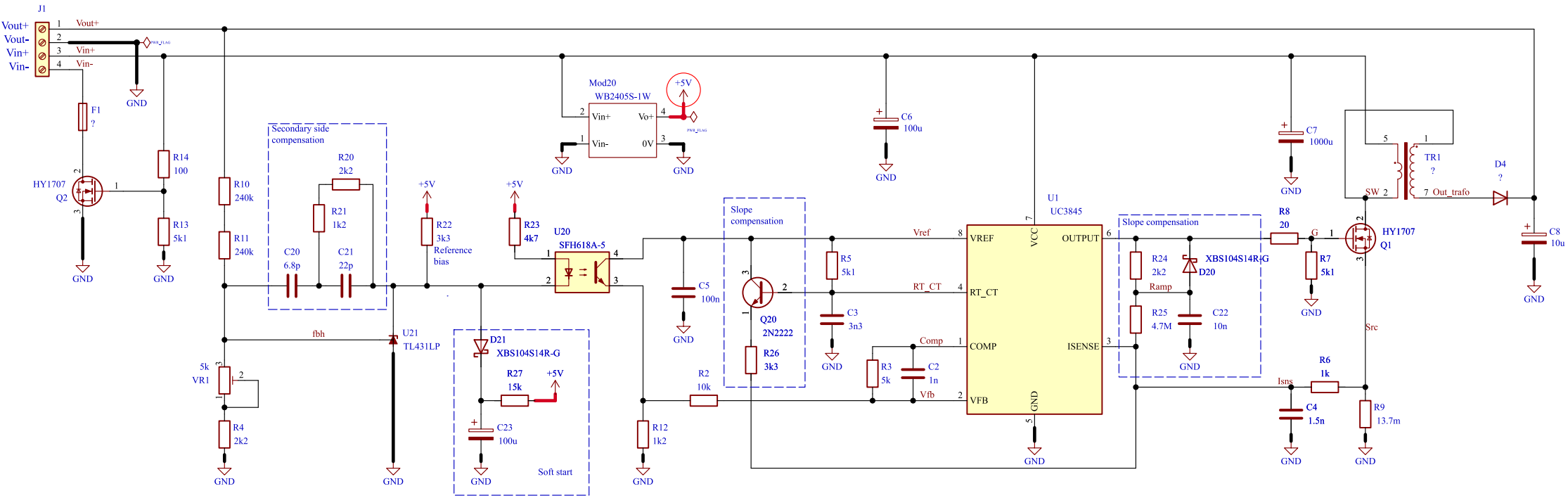
# [2] Block Diagram

MODULE INTERFACE:  
Vin: 9 V to 24 V according  
to AliExpress product page  
Vout: 200 V to 450 V,  
adjustable through VR1



|  |                                               |  |                                       |                 |                                            |                      |
|--|-----------------------------------------------|--|---------------------------------------|-----------------|--------------------------------------------|----------------------|
|  | Comments:<br>AliExpress module example (link) |  | Company:<br>FR                        |                 | Variant:<br>RELEASED                       | Git Hash:<br>f8fc805 |
|  |                                               |  | Board Name:<br>Module                 |                 | Project Name:<br>Isolated step-up 12:450 V |                      |
|  | Sheet Title:<br>Block Diagram                 |  | File Name:<br>Block Diagram.kicad_sch | Designer:<br>FR | Date:<br>2025-12-20                        | Revision:<br>1.0.0   |
|  | Sheet Path:<br>/Block Diagram/                |  |                                       | Reviewer:<br>NA | Size:<br>A4                                | Sheet:<br>2 of 4     |

[3] Schematic



- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

Estimated specs for unidentified / custom parts:  
- F1: ~5 A  
- D4: ~500 mA x 600 V  
- R9: ~13.7 mΩ (110 mV @ 8.04 A)  
- TR1: ~24:1 turns ratio, 10 μH : 5.76 mH

|  |                                          |                                   |                       |                     |                                            |                      |
|--|------------------------------------------|-----------------------------------|-----------------------|---------------------|--------------------------------------------|----------------------|
|  | Comments:                                |                                   | Company:<br>FR        |                     | Variant:<br>RELEASED                       | Git Hash:<br>f8fc805 |
|  |                                          |                                   | Board Name:<br>Module |                     | Project Name:<br>Isolated step-up 12:450 V |                      |
|  | Sheet Title:<br>Schematic                | File Name:<br>Schematic.kicad_sch | Designer:<br>FR       | Date:<br>2025-12-20 | Revision:<br>1.0.0                         |                      |
|  | Sheet Path:<br>/Block Diagram/Schematic/ |                                   | Reviewer:<br>NA       | Size:<br>A3         | Sheet:<br>3 of 4                           |                      |

# [4] Revision History

## Version 1.0.0 - 2026-01-24

### Change description

Cloned non-isolated design revision 1.1.0 as a working base.

Updated R10 and R11 to 240 kΩ (cloned design incorrectly used 140 kΩ).

Updated U1 to UC3845 (cloned design incorrectly used UC3843).

Reworked feedback loop to include opto-coupler while keeping the same reference on both sides to facilitate testing / reuse of original board.

Added two options of slope compensation / ramp injection (from U1 RT\_CT pin and from U1 OUTPUT pin).

Adjusted R2 to 10 kΩ, R3 to 5 kΩ and C2 to 1 nF to improve behaviour (output swaps from bursts to 0 % duty periodically).

Added soft-start (D21, R27 and C23).

Increased C5 to 100 nF.

Adjusted C4 to 1.5 nF and R6 to 1 kΩ (similar time constant) to facilitate ramp injection.

Updated parts placement where suitable to reuse existing layout and added an auxiliary perf-board to accomodate the new parts.

## Version 1.0.1 - 2025-12-21

|  |                                   |                                          |                 |                                            |                      |
|--|-----------------------------------|------------------------------------------|-----------------|--------------------------------------------|----------------------|
|  | Comments:                         | Company:<br>FR                           |                 | Variant:<br>RELEASED                       | Git Hash:<br>f8fc805 |
|  |                                   | Board Name:<br>Module                    |                 | Project Name:<br>Isolated step-up 12:450 V |                      |
|  | Sheet Title:<br>Revision History  | File Name:<br>Revision History.kicad_sch | Designer:<br>FR | Date:<br>2025-12-20                        | Revision:<br>1.0.0   |
|  | Sheet Path:<br>/Revision History/ |                                          | Reviewer:<br>NA | Size:<br>A4                                | Sheet:<br>4 of 4     |